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Finding the Cracks: Improving LLMs Reasoning with Paraphrastic Probing and Consistency Verification

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Abstract

Large language models have demonstrated impressive performance across a variety of reasoning tasks. However, their problem-solving ability often declines on more complex tasks due to hallucinations and the accumulation of errors within these intermediate steps. Recent work has introduced the notion of *critical tokens*–tokens in the reasoning process that exert significant influence on subsequent steps. Prior studies suggest that replacing critical tokens can refine reasoning trajectories. Nonetheless, reliably identifying and exploiting critical tokens remains challenging. To address this, we propose the **P**araphrastic **P**robing and **C**onsistency **V**erification (**PPCV**) framework. PPCV operates in two stages. In the first stage, we roll out an initial reasoning path from the original question and then concatenate paraphrased versions of the question with this reasoning path. And we identify critical tokens based on mismatches between the predicted top-1 token and the expected token in the reasoning path. A criterion is employed to confirm the final critical token. In the second stage, we substitute critical tokens with candidate alternatives and roll out new reasoning paths for both the original and paraphrased questions. The final answer is determined by checking the consistency of outputs across these parallel reasoning processes. We evaluate PPCV on mainstream LLMs across multiple benchmarks. Extensive experiments demonstrate PPCV substantially enhances the reasoning performance of LLMs compared to baselines.

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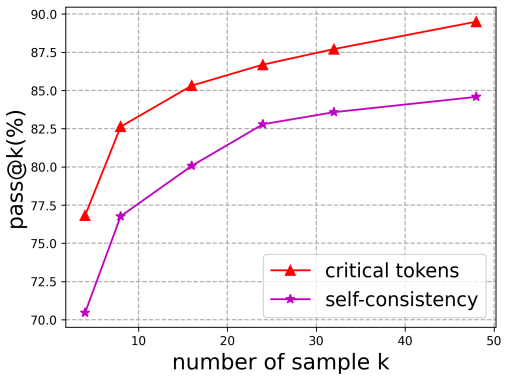

Figure 1. Comparison of the effects of critical tokens and Self-Consistency on the reasoning performance of LLMs, evaluated on samples from the GSM8K training data.

1. Introduction

The emergence of large language models (LLMs) (Brown et al., 2020; Grattafiori et al., 2024; Achiam et al., 2023; Yang et al., 2025a) has astonished the AI community with their remarkable capabilities across a wide range of reasoning tasks, including mathematical problem solving, programming, and planning. By generating intermediate reasoning steps through techniques such as chain-of-thought prompting (Wei et al., 2022; Kojima et al., 2022; Zhang et al., 2022; Sprague et al., 2025), LLMs can emulate human-like reasoning processes and achieve strong performance on diverse reasoning benchmarks. Despite their success, the problem-solving ability of LLMs often declines on complex reasoning tasks due to hallucinations and the accumulation of errors in intermediate steps (Lightman et al., 2023; Ling et al., 2023; Bubeck et al., 2023). To mitigate this issue, prior works (Wang et al., 2024; Yuan et al., 2025; Chen et al., 2024a; Chen & Li, 2024) have proposed the inference-time optimization paradigm, which evaluates and refines reasoning trajectories through self-correction by the LLM itself or with feedback from external verifiers such as process reward models (PRMs) (Uesato et al., 2022). However, existing studies (Yang et al., 2025b; Tyen et al., 2024; Stechly et al., 2025) suggest that LLMs struggle to reliably identify errors in their reasoning and often fail to correct previous outputs without external guidance. Moreover, obtaining high-quality, stepwise process supervision for training ex-

Case study on the effects of question paraphrasing

Original question: In a card game, you are awarded 10 points if you win one round. While playing, Jane ends up with 60 points. If she lost 20 points, then how many rounds did she play? (ground truth answer: 8)

Response: To solve this problem, we need to follow these steps:

Step 1: Determine the total points Jane earned by subtracting the points she lost from her final score. Total points earned = Final score - Points lost. Total points earned = 60 – 20. Total points earned = 40.

Step 2: Since Jane earns 10 points for each round she wins, we can find the number of rounds she played by dividing the total points earned by the points earned per round. Number of rounds = Total points earned / Points earned per round. Number of rounds = 40 / 10. Number of rounds = 4.

Therefore, Jane played 4 rounds. ✗

Paraphrased question: At a bridge competition, every game won grants the player 10 points. Unfortunately, Jane started with an advantage but ended up losing 20 points and scoring only 60 points. If she had won x rounds and lost none, what would be the value of x? (ground truth answer: 8)

Response: To solve this problem, we need to follow these steps:

Step 1: Understand the given information. Jane started with an advantage but ended up losing 20 points. She scored a total of 60 points. Every game won grants the player 10 points.

Step 2: Determine the total points Jane would have had if she hadn't lost any points. Since she lost 20 points, we need to add 20 to her total score to find out how many points she would have had if she hadn't lost any points. Total points without loss = 60 + 20 = 80 points.

Step 3: Calculate the number of games Jane won. Since every game won grants the player 10 points, we can divide the total points without loss by 10 to find the number of games she won. Number of games won = Total points without loss / 10 = 8. The number of games she won is equal to 8. ✓

Figure 2. Case study illustrating that LLMs can refine intermediate incorrect reasoning segments (highlighted in red).

ternal verifiers is challenging and limits the practicality of these approaches (Feng et al., 2025).

Recent work has introduced the concept of *critical tokens* (Lin et al., 2025), which play pivotal roles in intermediate reasoning steps and exert strong influence on subsequent reasoning and final outcomes. We evaluate the criticality of a token based on the average accuracy of sampling rollouts generated from the partial reasoning trajectories truncated at the token's position. A token is identified as critical if it lies on an incorrect trajectory and satisfies the following conditions:

- The correctness score of rollouts initiated from the token is close to zero (e.g., below 5%);
- The correctness scores of rollouts initiated from subsequent tokens remain below a threshold (e.g., 10%).

Prior studies suggest that critical tokens often diverge from human-annotated error tokens, yet they induce more sustained degradation in reasoning quality than other tokens. Moreover, as illustrated in Figure 3, replacing critical tokens in an incorrect reasoning trajectory with suitable candidate tokens can correct subsequent steps and lead to the right answer in new rollouts. To quantitatively assess the effectiveness of critical tokens, we conduct an empirical study using LLMs such as Llama-3.1-8B-Instruct (Grattafiori et al., 2024) on reasoning tasks. Specifically, we randomly sample 100 instances with incorrect reasoning steps from the GSM8K (Cobbe et al., 2021) training data. Following the criterion, we locate critical tokens through exhaustive search. We then truncate the reasoning path at the critical token, substitute it with alternative tokens, and roll out new reasoning paths. For example, as shown in Figure 3, the token "woman" is replaced with "remaining". We evaluate performance using Pass@k and compare against Self-Consistency (Wang et al., 2023), which also samples multiple reasoning paths. As shown in Figure 1, critical token replacement provides a clear advantage in correcting errors compared to plain sampling. Nonetheless, reliably identifying and leveraging critical tokens for reasoning remains a nontrivial challenge.

Case study on critical tokens

Question: There were 10000 spectators at a basketball match. 7000 of them were men. Of the remaining spectators, there were 5 times as many children as women. How many children were there? (ground truth answer: 2500)

Response:

To find the number of children, we need to first find the number of **women** and then multiply it by 5.

Step 1: Find the number of women. Total spectators = 10000. Men = 7000. Women = Total spectators – Men. Women = 10000 – 7000. Women = 3000.

Step 2: Find the number of children. There are 5 times as many children as women. Children = 5 * Women. Children = 5 * 3000. Children = 15000.

So, the number of children is 15000. ✗

Response(replacement of critical tokens):

To find the number of children, we need to first find the number of **remaining** spectators (excluding men) and then calculate the number of women and children.

Step 1: Find the number of remaining spectators (excluding men). Total spectators = 10000. Men = 7000. Remaining spectators = Total spectators – Men = 10000 - 7000= 3000

Step 2: Let's assume the number of women is 'x'. Since there are 5 times as many children as women, the number of children will be 5x.

Step 3: The total number of remaining spectators is the sum of women and children. We can write this as an equation: x + 5x = 3000. Combine like terms:\n6x = 3000. Divide both sides by 6: x = 3000 / 6. x = 500.

Step 4: Now that we have the number of women, we can find the number of children. Number of children = 5x= 5 * 500= 2500.

Therefore, the number of children is 2500. ✓

Figure 3. An example demonstrating how substitution of a critical token (red) with a candidate token (purple) modifies the reasoning path and produces the correct answer.

Recent studies (Zhou et al., 2024; Chen et al., 2024b), on *surface form*, the way questions, assumptions, and constraints are phrased, have revealed its subtle influence on the trajectory of intermediate reasoning steps, As shown in Figure 2. LLMs could adjust the intermediate steps under the paraphrased form of the question. This motivates us to explore the role of paraphrasing in the extraction and utilization of critical tokens for reasoning tasks. To this end, we propose the **P**araphrastic **P**robing and **C**onsistency **V**erification (**PPCV**) framework , a two-stage approach designed to leverage critical tokens to enhance the reasoning ability of LLMs. In the first stage, we probe critical tokens using paraphrased questions. Specifically, we first roll out the initial reasoning path from the original question, then concatenate paraphrased questions with this reasoning path. The resulting synthetic inputs are fed into the LLM to obtain token-level logits for each position in the reasoning path. Positions where the predicted top-1 token diverges from the expected token are flagged as potential pivotal points, as these positions are sensitive to paraphrased inputs and can trigger a pivot in the reasoning trajectory. Next, an empirical criterion is applied to determine the final critical token. In contrast to prior work (Lin et al., 2025), which depends on external models for identifying critical tokens with ambiguous criteria, our method introduces a self-contained mechanism that pinpoints critical tokens.

In the second stage, we leverage the extracted critical tokens to refine the initial reasoning path. Specifically, we select the top-K tokens (include critical token itself) at the critical token position and roll out new reasoning paths for both the original and paraphrased questions. Based on the empirical observation that trajectories leading to correct answers are robust to paraphrastic perturbations, we propose a paraphrase consistency mechanism. In contrast to Self-

Consistency (Wang et al., 2023), which relies on majority voting, our method selects the final answer by comparing outputs from paraphrased and original questions and choosing the one with the most consistent matches. To address potential ties across multiple answers, we further introduce similarity-weighted paraphrase consistency, which incorporates similarity scores between paraphrased and original questions when computing consistency.

Compared with self-correction (Wu et al., 2024; Miao et al., 2024) and PRM-based methods (Wang et al., 2024; Yuan et al., 2025), our framework exploits critical tokens to refine reasoning trajectories without requiring step-level error detection by the LLM itself or auxiliary models. We evaluate our method on mainstream LLMs across mathematical and commonsense reasoning benchmarks, demonstrating consistent improvements in reasoning performance. The contributions of the paper is summarized as follows:

- We propose a novel two-stage framework, **P**araphrastic **P**robing and **C**onsistency **V**erification (**PPCV**) designed to extract and leverage critical tokens to enhance the reasoning performance of LLMs.
- We show that critical tokens can more effectively correct erroneous reasoning trajectories than traditional sampling methods like Self-Consistency. Furthermore, our approach successfully extracts these tokens through paraphrastic probing, achieving improved final results via paraphrase consistency.
- We evaluate our method on mainstream LLMs across various reasoning tasks, including math and logic reasoning. Experimental results show significant performance improvements over baseline methods.

2. Related Work

Inference-Time Optimization for LLM reasoning. With the advent of chain-of-thought (CoT) prompting, LLMs have demonstrated strong reasoning capabilities by producing intermediate steps during inference. This success has motivated a growing body of work (Wu et al., 2025; Snell et al., 2024) on augmenting reasoning trajectories at test time to further improve performance. Existing approaches can be broadly categorized into search-based methods (Bi et al., 2025; Yao et al., 2023; Hao et al., 2023; Xie et al., 2023; Besta et al., 2024), such as Tree-of-Thoughts (Yao et al., 2023), and sampling-based methods (Wang et al., 2023; Xu et al., 2025; Wan et al., 2025; Ma et al., 2025), such as Self-Consistency (Wang et al., 2023). However, due to intrinsic hallucinations (Bubeck et al., 2023), LLMs often generate erroneous intermediate steps, which can ultimately lead to incorrect answers, especially on complex problems. This limitation highlights the need for inference-time optimization of reasoning processes.

To address this issue, one line of research (Yin et al., 2024; Chen et al., 2024a; Ling et al., 2023; Wu et al., 2024; Miao et al., 2024; Madaan et al., 2023) designs instructional prompts that guide LLMs to detect and refine their own mistakes. Despite its appeal, prior work has shown that the effectiveness of self-correction is limited in practice. Another line of work (Wang et al., 2024; Yuan et al., 2025; He et al., 2024; Havrilla et al., 2024) introduces external verifiers, such as process reward models (Snell et al., 2024), to identify and filter out error-prone reasoning steps. These methods typically require high-quality training data for the verifier, with data scarcity often mitigated through strategies such as Monte Carlo Tree Search (Guan et al., 2025; Qi et al., 2025; Li, 2025; Zhang et al., 2024). In addition, a recent line of decoding-based approaches (Xu et al., 2025; Ma et al., 2025) seeks to improve reasoning by dynamically adjusting the next-token prediction based on future trajectory probing. In contrast, our method refines reasoning by directly leveraging critical tokens, without relying on stepwise verification or external verifiers. This design underscores both the utility and universality of our approach.

Paraphrasing for LLMs. A growing number of work (Zhou et al., 2024; Chen et al., 2024b) has examined the impact of a problem's surface form on the reasoning ability of LLMs. Findings (Zhou et al., 2024; Chen et al., 2024b; Huang et al., 2025) suggest that even subtle modifications in phrasing can substantially affect both the reasoning process and the final outcome. Building on this observation, several methods (Yadav et al., 2024; Chen et al., 2024b) leverage beneficial paraphrasing to enhance LLM performance in tasks such as reasoning and intent classification. In addition, paraphrasing has been employed to assess model uncertainty (Gao et al., 2024; Tanneru et al., 2024), thereby enhancing the reliability and trustworthiness of LLM applications. In our work, we utilize paraphrasing as a principled tool to extract critical tokens and to aggregate answers.

3. Methodology

In this section, we present the two components of our framework in detail: paraphrastic probing and consistency verification. An detailed illustration of our framework is shown in Figure 4. We then discuss the proposed method and provide the complete algorithm.

3.1. Paraphrastic Probing

Previous findings (Zhou et al., 2024; Chen et al., 2024b; Huang et al., 2025) on the impact of a problem's surface form suggest that the quality of intermediate reasoning steps is influenced not only by the underlying mathematical relationships and logic, but also by how the problem is expressed. Notably, LLMs are sometimes able to solve a paraphrased version of a problem that they fail to solve in its